Cello Quantitative Research Plan

Cello Capital Management

February 18, 2015

Vision Statement

Build a comprehensive software framework (`Scale`) for the pricing, hedging and risk management of fixed-income instruments and, in particular, structured products arising in the housing and consumer finance markets. The library should be written in C++ with a clean object model that allows it to be exported to different languages such as C♯, Python and R. It should also be possible to export an object-oriented `Scale` interface to end-user platforms such as Microsoft Excel.

Benefits of Scale

- Micro-level specification of proprietary models
- Reduce low-level model building work and allow focus on more complex problems
- Platform for scaling the firm across asset classes and strategies
- Tangible repository for Cello IP
- Training ground for new Associates/Analysts in the Quantitative Research group

Required Investments

- **Time**: Building a multi-asset class fixed-income analytics library is a huge task
- **Quants**: For model building
- **Software Engineers**: Creating library interfaces, utilities
- **Process**: Enforcing good design, coding standards, code reviews, testing
- **Delivery System**: Easy access to model outputs, ability to perturb inputs and assumptions.

Challenges

- **Prolonged gestation period**: Several man-years of work required before `Scale` will be mature enough to supplant vendor systems such as YieldBook. Every effort will be made to employ robust open source code (where applicable) in order to decrease development time frames.
- **Robust implementation**: Institution of and rigorous adherence to a software development process is necessary to deliver a high-quality library
- **IT delivery system**: Distributed computing platform is required to allow library to be called simultaneously by multiple users.

Core Components of `Scale`: Phase I

- Yield Curve Builder (Libor/Swap Curve)
- 1-factor Hull-White Interest Rate Model
- 30-year Conventional Mortgage Rate Model
- Pass-through and IOS cash flow engine
- 30-year Fannie Mae Prepayment model
- Conventional House Price model
- Risk Engine (OAS, Durations etc.)
- Roll-rate Model for GSE Credit Risk Transfer Deals
- Utility functions (Dates, Splines, Optimizer/Solver etc)

Process and Methodology for `Scale`: Phase I

- Documentation
- Source control and software versioning
- One-step daily builds
- Bug database
- Unit testing
- Coding standards

To-Dos: Q1 2015

- Staffing plan: SC (50%), RM (100%), LC (75%), NW (25%)
- Productionize daily swap curve constructor for `Scale`. Load historical data.
- Define best practices for process and methodology
- Define project deliverables and time scales for IT delivery system